

to put in the cartridge, and instead of paper, you could think of just about anything for the 'ink' to be deposited on.

Although printing out skin is an interesting and useful idea in itself, one can easily extend the concept. How about printing out display screens? Printed circuit boards? MP3 players? Entire houses?

But we're getting ahead of ourselves. In what follows, we'll talk about 3D printing—instead of just one print head moving to and fro in one dimension, imagine several print heads capable of moving in all three dimensions, depositing anything you could imagine on any suitable surface. Or, a regular inkjet printer could print layer upon layer, thus making up a 3D object—for example, a bone or an internal organ.

What you have then, are machines capable of printing, well, just about anything. For example, instead of constructing an MP3 player part by part, incorporating the PCB, the plastic moulding, the hard disk and so on, you could just create a blueprint of the entire device and print it all out at once. That's the idea behind most of what follows—a plan being printed out.

Here's a glimpse at what is being printed now, and what will be printed in the near future; and all of it goes far beyond plain old paper and ink.

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Skin, Bone And Cartilage

So how does all this work? Like we mentioned, the inkjet printer is a marvel of technology because of the print nozzle, which spits out thousands of microdroplets. As an example, think about grafting skin onto a wound. A 2D inkjet printer will suffice for skin because the proteins shot out can be captured on a flat gel substrate layer. Experts enter the wound's dimensions into the printer to ensure a perfect fit. They then take skin cells from the patient's body and multiply them using biological techniques, and then print out a strip of skin that is tailor-made for the wound—and ready to sew on.

Such a printer has been developed at Manchester University in England; it only takes up the area of three filing cabinets. At the university's School of Materials, scientists have already successfully created skin, and think they will soon be able to create bone and cartilage as well.

Team leader Professor Brian Derby says, "It's not like printing a sheet of paper. We can print a few millimetres in depth and build it up layer-upon-layer until, in principle, we could produce bone fragments the size of a golf ball." Project leaders say the method could eventually be used to build an organ in a day!

The amazing thing here is that it's basically inkjet technology at work.



Inkjets For Just About Anything

Faxing Up Spare Parts

Think about the plight of an astronaut in space when something breaks. Where would he get a spare part from? Roger Spielman, a scientist for Boeing Canoga Park, has the answer: fax and print. One could 'fax' the 3D computer-aided design file to the International Space Station, where the data would be fed into a machine that assembles the required spare part from a bucket of powder. This, if you think about it, is not very far from faxing a physical object!

Printing Out A Heart

Tissue engineering combines the disciplines of cell biology, materials engineering, and biochemistry to create biological substitutes for human and animal tissue. At Clemson University in South Carolina, chemical engineer Thomas Boland had a novel idea in 2001 that brought the inkjet into tissue engineering. His method was deceptively simple: you could build up a 3D tissue by modifying the printers' output tray to drop down a little bit with each pass of the print head. Boland used his printers to create half a cat's heart—the heart actually beat in a petri dish!

Fuel Injection And Medicine Atomising

Since an inkjet sprays tiny droplets, one of the first uses that springs to mind is fuel injection. A regular injector does the same thing, atomising petrol (or diesel) into tiny droplets, but the smaller the droplets, the more completely they will vaporise and mix with air, making for more efficient fuel burning. An inkjet can therefore be part of extremely precisely controlled fuel injectors.

Think about medical inhalers, such as asthma medication inhalers. As with fuel injection, the task of creating the spray that these inhalers spit out is ideally suited to the inkjet. The problem is that conventional inhalers spray out a mist of drops that are too large, and also uneven. The inkjet can remedy this: digital aerosols, which are thermal print heads incorporated into inhalers that spit out perfectly uniform drops, could come to market as

early as the end of 2005. This method of injection is ideally suited to drugs that need to get to the bloodstream—the smaller the size of the droplets, the easier the absorption of the medicine into the bloodstream.

Digital aerosols are being developed by Australian company InJet Digital Aerosols. The inhalers will deliver drugs faster than do tablets or patches.

Flexible Plastic Screens

Several companies worldwide are in the race to commercialise flexible plastic-screen technology. Philips is currently using a four-headed inkjet with hundreds of piezoelectric nozzles to print OLEDs (Organic Light-Emitting Diodes) onto computer and television screens. You have probably heard of OLED screens—OLEDs glow when a current is applied. (Piezoelectric nozzles are one of the two commonly-used types of nozzle—the other is a thermal nozzle. A piezoelectric nozzle works when a current bends a piezoelectric crystal, forcing the fluid down and out of the nozzle. One of the properties of piezoelectric crystals is that they move or bend when a current is applied, and they release a current when bent.)

In Bristol, England, HP engineers are using inkjets to etch tiny liquid-crystal dots onto a plastic substrate—the dots themselves will be the pixels. Electrodes in the plastic, which is bendable, turn the crystals on and off, making for a full-colour display that would, at some point, compare with printed material in terms of resolution.

Circuit Boards

As was proved by Japanese engineers at Seiko-Epson, printing the innards of a gadget is possible using inkjets. In November 2003, they printed a circuit board consisting of 20 layers, and only 200 microns thick. How? They simply swapped copper for conductive inks and silicon for insulating inks. A piezoelectric print head with microscopic nozzles was used for the process. Seiko-Epson is confident that these thin boards will be used in gadgets as early as 2007!